

SIMULATION IDEAS

For each observation $i \in [n]$, denote by its tensor covariate $\mathcal{X}_i \in \mathbb{R}^{p_1 \times \dots \times p_D}$, one-way covariates $\mathbf{z}_i \in \mathbb{R}^{p_0}$, and two responses $y_{ij} \in \mathbb{R}$ where $j = 1, 2$ represents two eyes. Let $t \in [0, 1]$ be the observation time, $\mathcal{A}(t) = (\alpha_{j_1, \dots, j_d}(t)) \in \mathbb{R}^{p_1 \times \dots \times p_D}$ be the unknown tensor coefficient function, and $\boldsymbol{\beta} \in \mathbb{R}^{p_0}$ be the coefficients. The model is assumed to be

$$y_{ij} = \langle \mathcal{X}_{ij}, \mathcal{A}(t_i) \rangle + \mathbf{z}_i^\top \boldsymbol{\beta} + \epsilon_{ij} = \sum_{s=1}^S \left\langle \mathcal{X}_{ij}^{(s)}, \mathcal{A}^{(s)}(t_i) \right\rangle + \mathbf{z}_i^\top \boldsymbol{\beta} + \epsilon_{ij} \quad (1)$$

by a rank- R CP decomposition, we can reorganize it in a matrix form

$$\approx \sum_{s=1}^S \sum_{r=1}^R x_{ijrs}^* \left\langle \mathcal{U}_r^{(s)}, \mathcal{A}^{(s)}(t_i) \right\rangle + \mathbf{z}_i^\top \boldsymbol{\beta} + \epsilon_{ij} = \langle \mathbf{X}_{ij}^*, \mathbf{A}(t_i) \rangle + \mathbf{z}_i^\top \boldsymbol{\beta} + \epsilon_{ij},$$

where matrix $\mathbf{X}_{ij}^* := (x_{ijrs}^*) \in \mathbb{R}^{R \times S}$, and $\mathbf{A}(t) := (\alpha_{rs}(t)) \in \mathbb{R}^{R \times S}$ with $\alpha_{rs}(t) := \left\langle \mathcal{U}_r^{(s)}, \mathcal{A}^{(s)}(t) \right\rangle$. Within one patient, the eyes are correlated by

$$\begin{bmatrix} \epsilon_{i1} \\ \epsilon_{i2} \end{bmatrix} \sim \mathbf{N} \left(\mathbf{0}, \sigma^2(t) \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix} \right).$$

1 Paired Simulation for Case I

DGP. The tensor size is set to be $D = 2$, $p_1 = p_2 = 60$, and split by $p'_1 = p'_2 = 12$ such that by $S^{(1)} = S^{(2)} = 5$. Set the rank $R = 4$, and the s -th block tensor for j -th eye of i -th patient is generated by:

$$\mathcal{X}_{ij}^{(s)} = \sum_{r=1}^R x_{ijrs}^* \mathbf{x}_r^{(s1)} \circ \mathbf{x}_r^{(s2)}, \quad \text{where } x_{ijrs}^* \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1).$$

and any $\mathbf{x}_r^{(sd)}$ and $\mathbf{x}_{r'}^{(sd)}$ are orthonormal and obtained from the QR decomposition of a random matrix. The coefficient function is set to be $\mathcal{A}^{(s)}(t) = \sum_{r=1}^R \alpha_{rs}(t) \mathbf{x}_r^{(s1)} \circ \mathbf{x}_r^{(s2)}$ where $\alpha_{rs}(t)$ has 4 choices:

$$\alpha_{1rs}(t) = 5\sqrt{\frac{rs}{RS}}(t - 0.2)^2, \quad \alpha_{2rs}(t) = \sqrt{\frac{rs}{RS}} \left(e^{-(3t-1)^2} - 0.75 \right), \quad \alpha_{3rs}(t) = \sqrt{\frac{rs}{RS}} \sin(2\pi(t - 0.5)), \quad (2)$$

$$\alpha_{4rs}(t) = \sqrt{\frac{rs}{RS}} \left[0.45 \cdot 5(t - 0.2)^2 + 0.35(e^{-(3t-1)^2} - 0.75) + 0.20 \sin(2\pi(t - 0.5)) \right]. \quad (3)$$

The one-way vector $\mathbf{Z} \in \mathbb{R}^{n \times p_0}$ is Gaussian ensemble matrix (iid entries) and $\boldsymbol{\beta} = (2.0, 1.0, -1.0, 0.5)^\top$. For now, we set $\sigma^2(t) \equiv 1$ as a toy model. The responses are generated by model eq(1).

Estimation Steps. (1) We treat the paired data as iid, and get initial estimate $\hat{\mathbf{A}}^\dagger(t)$ and $\hat{\boldsymbol{\beta}}^\dagger$. (2) Then estimate covariance structure $\hat{\sigma}^2(t)$ by eq(15)-(16) in AOAS draft. (3) Get final estimates $\hat{\mathbf{A}}^*(t)$ and $\hat{\boldsymbol{\beta}}^*$ from eq(14)-(15) of AOAS draft.

Simulation Results. We vary $n = 1000/2000$, coefficients $\alpha_{1rs}(t), \dots, \alpha_{4rs}(t)$, and correlation $\rho = 0/0.3/0.6/0.9$. We report the error of both initial estimates $(\hat{\mathbf{A}}^\dagger, \hat{\boldsymbol{\beta}}^\dagger)$ and final estimates $(\hat{\mathbf{A}}^*, \hat{\boldsymbol{\beta}}^*)$, also the error of $(\hat{\sigma}^2, \hat{\rho})$.

Tables. All values in the following tables are reported in units of $\times 10^{-2}$; within each row, the better value between $\hat{\mathbf{A}}^\dagger(t)$ or $\hat{\mathbf{A}}^*(t)$ is highlighted.

Table 1: $\mathbf{A}(t)$ MIAE and β MAE: phase 1 vs phase 3.

function	n	ρ	$\hat{\mathbf{A}}^\dagger(t)$	$\hat{\mathbf{A}}^*(t)$	$\hat{\beta}^\dagger$	$\hat{\beta}^*$
$\mathbf{A}_1(t)$	1000	0.0	4.35 ($\pm$ 0.26)	4.35 ($\pm$ 0.26)	2.07 ($\pm$ 0.78)	2.07 ($\pm$ 0.78)
		0.3	4.55 ($\pm$ 0.33)	4.36 ($\pm$ 0.31)	2.03 ($\pm$ 0.89)	2.06 ($\pm$ 0.89)
		0.6	4.53 ($\pm$ 0.33)	3.69 ($\pm$ 0.26)	2.23 ($\pm$ 0.98)	2.28 ($\pm$ 0.99)
		0.9	4.48 ($\pm$ 0.34)	2.07 ($\pm$ 0.14)	2.40 ($\pm$ 1.07)	2.47 ($\pm$ 1.07)
	2000	0.0	3.12 ($\pm$ 0.20)	3.12 ($\pm$ 0.20)	1.43 ($\pm$ 0.60)	1.43 ($\pm$ 0.60)
		0.3	3.09 ($\pm$ 0.20)	2.95 ($\pm$ 0.20)	1.83 ($\pm$ 0.82)	1.82 ($\pm$ 0.81)
		0.6	3.08 ($\pm$ 0.20)	2.49 ($\pm$ 0.18)	2.03 ($\pm$ 0.91)	2.01 ($\pm$ 0.88)
		0.9	3.07 ($\pm$ 0.20)	1.40 ($\pm$ 0.10)	2.20 ($\pm$ 0.99)	2.16 ($\pm$ 0.94)
$\mathbf{A}_2(t)$	1000	0.0	4.40 ($\pm$ 0.28)	4.40 ($\pm$ 0.28)	2.06 ($\pm$ 0.80)	2.06 ($\pm$ 0.80)
		0.3	4.60 ($\pm$ 0.36)	4.42 ($\pm$ 0.35)	2.02 ($\pm$ 0.90)	2.05 ($\pm$ 0.90)
		0.6	4.57 ($\pm$ 0.36)	3.75 ($\pm$ 0.30)	2.21 ($\pm$ 1.00)	2.27 ($\pm$ 1.00)
		0.9	4.51 ($\pm$ 0.36)	2.17 ($\pm$ 0.18)	2.39 ($\pm$ 1.08)	2.46 ($\pm$ 1.07)
	2000	0.0	3.16 ($\pm$ 0.21)	3.16 ($\pm$ 0.21)	1.43 ($\pm$ 0.61)	1.44 ($\pm$ 0.61)
		0.3	3.10 ($\pm$ 0.20)	2.97 ($\pm$ 0.21)	1.84 ($\pm$ 0.82)	1.83 ($\pm$ 0.80)
		0.6	3.10 ($\pm$ 0.20)	2.52 ($\pm$ 0.19)	2.03 ($\pm$ 0.90)	2.01 ($\pm$ 0.87)
		0.9	3.10 ($\pm$ 0.20)	1.48 ($\pm$ 0.12)	2.20 ($\pm$ 0.98)	2.17 ($\pm$ 0.93)
$\mathbf{A}_3(t)$	1000	0.0	4.58 ($\pm$ 0.29)	4.58 ($\pm$ 0.29)	2.10 ($\pm$ 0.83)	2.10 ($\pm$ 0.83)
		0.3	4.78 ($\pm$ 0.32)	4.59 ($\pm$ 0.27)	2.00 ($\pm$ 0.88)	2.03 ($\pm$ 0.88)
		0.6	4.76 ($\pm$ 0.34)	3.95 ($\pm$ 0.22)	2.20 ($\pm$ 0.97)	2.25 ($\pm$ 0.98)
		0.9	4.72 ($\pm$ 0.35)	2.50 ($\pm$ 0.12)	2.36 ($\pm$ 1.06)	2.43 ($\pm$ 1.07)
	2000	0.0	3.45 ($\pm$ 0.21)	3.45 ($\pm$ 0.21)	1.45 ($\pm$ 0.59)	1.45 ($\pm$ 0.59)
		0.3	3.48 ($\pm$ 0.20)	3.36 ($\pm$ 0.20)	1.84 ($\pm$ 0.79)	1.83 ($\pm$ 0.78)
		0.6	3.48 ($\pm$ 0.21)	2.95 ($\pm$ 0.18)	2.03 ($\pm$ 0.88)	2.01 ($\pm$ 0.85)
		0.9	3.47 ($\pm$ 0.21)	2.04 ($\pm$ 0.11)	2.20 ($\pm$ 0.96)	2.16 ($\pm$ 0.91)
$\mathbf{A}_4(t)$	1000	0.0	4.99 ($\pm$ 0.41)	4.99 ($\pm$ 0.41)	2.02 ($\pm$ 0.86)	2.02 ($\pm$ 0.86)
		0.3	5.15 ($\pm$ 0.40)	4.99 ($\pm$ 0.37)	2.02 ($\pm$ 0.93)	2.04 ($\pm$ 0.94)
		0.6	5.11 ($\pm$ 0.41)	4.41 ($\pm$ 0.31)	2.22 ($\pm$ 1.02)	2.26 ($\pm$ 1.02)
		0.9	5.05 ($\pm$ 0.40)	3.16 ($\pm$ 0.21)	2.39 ($\pm$ 1.11)	2.46 ($\pm$ 1.09)
	2000	0.0	3.84 ($\pm$ 0.28)	3.84 ($\pm$ 0.28)	1.48 ($\pm$ 0.63)	1.49 ($\pm$ 0.64)
		0.3	3.89 ($\pm$ 0.20)	3.77 ($\pm$ 0.21)	1.82 ($\pm$ 0.78)	1.81 ($\pm$ 0.77)
		0.6	3.90 ($\pm$ 0.19)	3.43 ($\pm$ 0.20)	2.01 ($\pm$ 0.87)	1.99 ($\pm$ 0.84)
		0.9	3.90 ($\pm$ 0.21)	2.76 ($\pm$ 0.13)	2.18 ($\pm$ 0.95)	2.14 ($\pm$ 0.90)

Table 2: $\mathbf{A}(t)$ RMISE and β RMSE: phase 1 vs phase 3.

function	n	ρ	$\hat{\mathbf{A}}^\dagger(t)$	$\hat{\mathbf{A}}^*(t)$	$\hat{\beta}^\dagger$	$\hat{\beta}^*$
$\mathbf{A}_1(t)$	1000	0.0	5.68 ($\pm$ 0.32)	5.68 ($\pm$ 0.32)	2.27 ($\pm$ 0.87)	2.27 ($\pm$ 0.87)
		0.3	5.99 ($\pm$ 0.47)	5.74 ($\pm$ 0.42)	2.39 ($\pm$ 1.07)	2.41 ($\pm$ 1.06)
		0.6	5.95 ($\pm$ 0.48)	4.87 ($\pm$ 0.36)	2.63 ($\pm$ 1.19)	2.67 ($\pm$ 1.18)
		0.9	5.88 ($\pm$ 0.49)	2.75 ($\pm$ 0.20)	2.83 ($\pm$ 1.31)	2.89 ($\pm$ 1.28)
	2000	0.0	4.09 ($\pm$ 0.25)	4.09 ($\pm$ 0.24)	1.60 ($\pm$ 0.63)	1.60 ($\pm$ 0.64)
		0.3	4.02 ($\pm$ 0.28)	3.85 ($\pm$ 0.28)	2.07 ($\pm$ 0.91)	2.06 ($\pm$ 0.90)
		0.6	4.01 ($\pm$ 0.28)	3.26 ($\pm$ 0.25)	2.29 ($\pm$ 1.01)	2.27 ($\pm$ 0.98)
		0.9	4.01 ($\pm$ 0.27)	1.84 ($\pm$ 0.14)	2.49 ($\pm$ 1.10)	2.45 ($\pm$ 1.06)
$\mathbf{A}_2(t)$	1000	0.0	5.74 ($\pm$ 0.34)	5.74 ($\pm$ 0.34)	2.27 ($\pm$ 0.88)	2.27 ($\pm$ 0.88)
		0.3	6.02 ($\pm$ 0.47)	5.78 ($\pm$ 0.44)	2.37 ($\pm$ 1.08)	2.39 ($\pm$ 1.06)
		0.6	5.98 ($\pm$ 0.49)	4.91 ($\pm$ 0.38)	2.61 ($\pm$ 1.20)	2.65 ($\pm$ 1.18)
		0.9	5.91 ($\pm$ 0.50)	2.82 ($\pm$ 0.22)	2.82 ($\pm$ 1.31)	2.88 ($\pm$ 1.27)
	2000	0.0	4.13 ($\pm$ 0.26)	4.13 ($\pm$ 0.26)	1.60 ($\pm$ 0.64)	1.60 ($\pm$ 0.64)
		0.3	4.04 ($\pm$ 0.29)	3.86 ($\pm$ 0.29)	2.07 ($\pm$ 0.91)	2.06 ($\pm$ 0.90)
		0.6	4.03 ($\pm$ 0.28)	3.28 ($\pm$ 0.26)	2.30 ($\pm$ 1.01)	2.27 ($\pm$ 0.99)
		0.9	4.04 ($\pm$ 0.27)	1.91 ($\pm$ 0.15)	2.50 ($\pm$ 1.10)	2.46 ($\pm$ 1.06)
$\mathbf{A}_3(t)$	1000	0.0	5.93 ($\pm$ 0.34)	5.93 ($\pm$ 0.34)	2.31 ($\pm$ 0.93)	2.31 ($\pm$ 0.93)
		0.3	6.23 ($\pm$ 0.46)	5.99 ($\pm$ 0.40)	2.35 ($\pm$ 1.07)	2.38 ($\pm$ 1.06)
		0.6	6.19 ($\pm$ 0.48)	5.16 ($\pm$ 0.32)	2.59 ($\pm$ 1.19)	2.63 ($\pm$ 1.17)
		0.9	6.12 ($\pm$ 0.50)	3.25 ($\pm$ 0.17)	2.79 ($\pm$ 1.30)	2.85 ($\pm$ 1.28)
	2000	0.0	4.45 ($\pm$ 0.24)	4.45 ($\pm$ 0.24)	1.61 ($\pm$ 0.63)	1.61 ($\pm$ 0.63)
		0.3	4.45 ($\pm$ 0.27)	4.30 ($\pm$ 0.26)	2.06 ($\pm$ 0.88)	2.05 ($\pm$ 0.87)
		0.6	4.44 ($\pm$ 0.28)	3.78 ($\pm$ 0.23)	2.29 ($\pm$ 0.98)	2.26 ($\pm$ 0.95)
		0.9	4.43 ($\pm$ 0.28)	2.62 ($\pm$ 0.13)	2.49 ($\pm$ 1.07)	2.45 ($\pm$ 1.03)
$\mathbf{A}_4(t)$	1000	0.0	6.37 ($\pm$ 0.48)	6.37 ($\pm$ 0.47)	2.25 ($\pm$ 0.94)	2.24 ($\pm$ 0.94)
		0.3	6.60 ($\pm$ 0.51)	6.38 ($\pm$ 0.46)	2.37 ($\pm$ 1.09)	2.39 ($\pm$ 1.07)
		0.6	6.56 ($\pm$ 0.52)	5.60 ($\pm$ 0.38)	2.60 ($\pm$ 1.21)	2.64 ($\pm$ 1.18)
		0.9	6.49 ($\pm$ 0.51)	3.93 ($\pm$ 0.24)	2.81 ($\pm$ 1.33)	2.87 ($\pm$ 1.28)
	2000	0.0	4.87 ($\pm$ 0.33)	4.87 ($\pm$ 0.33)	1.64 ($\pm$ 0.68)	1.65 ($\pm$ 0.68)
		0.3	4.90 ($\pm$ 0.27)	4.74 ($\pm$ 0.27)	2.04 ($\pm$ 0.89)	2.03 ($\pm$ 0.88)
		0.6	4.90 ($\pm$ 0.27)	4.28 ($\pm$ 0.24)	2.26 ($\pm$ 0.98)	2.24 ($\pm$ 0.96)
		0.9	4.91 ($\pm$ 0.28)	3.35 ($\pm$ 0.15)	2.46 ($\pm$ 1.07)	2.43 ($\pm$ 1.03)

- (i) When $\rho = 0$, there is no difference between iid estimation $(\hat{\mathbf{A}}^\dagger(t), \hat{\beta}^\dagger)$ and paired estimation $(\hat{\mathbf{A}}^*(t), \hat{\beta}^*)$.
- (ii) The larger the ρ is, the better the paired estimation $\hat{\mathbf{A}}^*(t)$ is than iid $\hat{\mathbf{A}}^\dagger(t)$.
- (iii) The larger the ρ is, the harder to estimate β for both paired estimation $\hat{\beta}^*$ and iid estimation $\hat{\beta}^\dagger$.
- (iv) All estimation errors decrease as n increases.

Table 3: Covariance estimation errors.

function	n	MIAE of $\hat{\sigma}^2(t)$				MAE of $\hat{\rho}$			
		$\rho = 0$	$\rho = 0.3$	$\rho = 0.6$	$\rho = 0.9$	$\rho = 0$	$\rho = 0.3$	$\rho = 0.6$	$\rho = 0.9$
$\mathbf{A}_1(t)$	1000	6.00 ($\pm$ 3.43)	6.16 ($\pm$ 3.35)	6.11 ($\pm$ 3.67)	6.22 ($\pm$ 4.09)	1.67 ($\pm$ 1.44)	3.38 ($\pm$ 2.24)	3.08 ($\pm$ 2.20)	4.15 ($\pm$ 0.97)
	2000	2.94 ($\pm$ 1.77)	3.11 ($\pm$ 1.89)	3.32 ($\pm$ 2.27)	3.62 ($\pm$ 2.62)	2.34 ($\pm$ 1.56)	2.02 ($\pm$ 1.59)	2.02 ($\pm$ 1.44)	2.38 ($\pm$ 0.54)
$\mathbf{A}_2(t)$	1000	5.93 ($\pm$ 3.43)	6.09 ($\pm$ 3.34)	6.03 ($\pm$ 3.67)	6.16 ($\pm$ 4.09)	1.65 ($\pm$ 1.45)	3.40 ($\pm$ 2.22)	3.07 ($\pm$ 2.26)	4.20 ($\pm$ 0.99)
	2000	2.88 ($\pm$ 1.76)	3.04 ($\pm$ 1.89)	3.25 ($\pm$ 2.27)	3.57 ($\pm$ 2.60)	2.34 ($\pm$ 1.55)	2.04 ($\pm$ 1.60)	2.07 ($\pm$ 1.43)	2.42 ($\pm$ 0.54)
$\mathbf{A}_3(t)$	1000	5.61 ($\pm$ 3.33)	5.74 ($\pm$ 3.28)	5.70 ($\pm$ 3.60)	5.89 ($\pm$ 3.95)	1.67 ($\pm$ 1.48)	3.45 ($\pm$ 2.25)	3.29 ($\pm$ 2.28)	4.58 ($\pm$ 1.02)
	2000	2.51 ($\pm$ 1.69)	2.68 ($\pm$ 1.85)	2.97 ($\pm$ 2.14)	3.34 ($\pm$ 2.42)	2.31 ($\pm$ 1.62)	2.03 ($\pm$ 1.63)	2.24 ($\pm$ 1.50)	2.84 ($\pm$ 0.54)
$\mathbf{A}_4(t)$	1000	5.09 ($\pm$ 3.16)	5.17 ($\pm$ 3.10)	5.14 ($\pm$ 3.42)	5.41 ($\pm$ 3.75)	1.71 ($\pm$ 1.56)	3.44 ($\pm$ 2.23)	3.52 ($\pm$ 2.34)	5.15 ($\pm$ 0.96)
	2000	1.98 ($\pm$ 1.49)	2.12 ($\pm$ 1.69)	2.50 ($\pm$ 1.89)	2.89 ($\pm$ 2.19)	2.37 ($\pm$ 1.65)	2.14 ($\pm$ 1.76)	2.74 ($\pm$ 1.49)	3.52 ($\pm$ 0.56)

For covariance estimation, both $\hat{\sigma}^2$ and $\hat{\rho}$ are estimated with reasonable accuracy. And

- The larger the ρ is, the harder to estimate it.
- The larger the n is, the smaller the error is.

2 Paired Simulation for Case II

DGP. The tensor size is set to be $D = 3$, $p_1 = p_2 = p_3 = 40$, and $S = 64$. We split the tensor by $S^{(1)} = S^{(2)} = S^{(3)} = 10$ and $p'_1 = p'_2 = p'_3 = 4$. Set the rank $R = 5$, and covariate tensor:

$$\mathcal{X}_{ij}^{(s)} = \sum_{r=1}^R x_{ijrs}^* \mathbf{x}_r^{(s1)} \circ \mathbf{x}_r^{(s2)} \circ \mathbf{x}_r^{(s3)}, \text{ where } x_{ijrs}^* \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1)$$

and any $\mathbf{x}_r^{(sd)}$ and $\mathbf{x}_{r'}^{(sd)}$ are orthonormal and obtained from the QR decomposition of a random matrix. The coefficient function is set to be $\mathcal{A}^{(s)}(t) = \sum_{r=1}^R \alpha_{rs}(t) \mathbf{x}_r^{(s1)} \circ \mathbf{x}_r^{(s2)} \circ \mathbf{x}_r^{(s3)}$ where $\alpha_{rs}(t)$ has 4 choices in Eq(2) and (3). Everything else is same as Case I.

Simulation Results. We vary $n = 500/1000/2000$, coefficients $\mathcal{A}_1(t), \dots, \mathcal{A}_4(t)$, and correlation $\rho = 0/0.3/0.6/0.9$. We report the error of both initial estimates $(\hat{\mathbf{A}}^\dagger, \hat{\beta}^\dagger)$ and final estimates $(\hat{\mathbf{A}}^*, \hat{\beta}^*)$, also the error of $(\hat{\sigma}^2, \hat{\rho})$.

Simulations to be done...